

Baisakhi Banik

✉ baisakhibanik14@gmail.com

☎ +91 9800707159

📞 9800707159

🌐 <https://www.researchgate.net/profile/Baisakhi-Banik>



Personal Details

- 📌 **Name:** Baisakhi Banik
- 📌 **Father's Name:** Bapi Banik
- 📌 **Mother's Name:** Dipa Banik
- 📌 **D.O.B.:** 22.04.1996
- 📌 **Address:** Arabinda Road, Habra, P.O-Habra, P.S.-Habra, Dist- North 24 Parganas, Pin-743263, West Bengal, India.

Current Position

- 📌 Senior Research Fellow, Department of Mathematics, Indian Institute of Engineering Science and Technology, Shibpur, India under the supervision of Dr. Shariful Alam.

Education

- 2021 – Till date 📌 **Ph.D.** (pursuing and about to submit the thesis) from Indian Institute of Engineering Science and Technology, Shibpur.
Broad Area of Research: *Decision Making under Uncertain Domain.*
- 2017 – 2019 📌 **M.Sc** from Indian Institute of Engineering Science and Technology, Shibpur.
Obtained Marks/C.G.P.A: *1695 out of 2000 / 9.02.*
Subjects/ Discipline: *Applied Mathematics.*
- 2014 – 2017 📌 **B.Sc Honours** from Barasat Government College (West Bengal State University).
Obtained Marks/Percentage: *578 out of 800 / 72.25%.*
Subjects/ Discipline: *Major: Mathematics, Minor: Physics, Chemistry.*
- 2014 📌 **Higher Secondary** from West Bengal Council of Higher Secondary Education.
Obtained Marks/Percentage: *410 out of 500 / 82.00%.*
Subjects/ Discipline: *Bengali, English, Mathematics, Physics, Chemistry, Biology.*
- 2012 📌 **Secondary** from West Bengal Board of Secondary Education.
Obtained Marks/Percentage: *617 out of 700 / 88.14%.*
Subjects/ Discipline: *Bengali, English, Mathematics, Physical Science, Life Science, History, Geography.*

Area of Interest

- 📌 Fuzzy sets, Intuitionistic sets, Neutrosophic Sets.
- 📌 Decision Making.
- 📌 Inventory Management.
- 📌 Supply Chain Management.

Employment History

Feb, 2024	■ Supervisor. Integrated Child Development Scheme, Department of Women and Child Development and Social Welfare, Government of West Bengal. (not availed)
Nov, 2020 – Mar, 2021	■ Guest Faculty. Narula Institute of Technology, Agarpara, Department of Basic Science and Humanity.
Nov, 2019 – Aug, 2020	■ Assistant Professor. Techno Engineering College, Banipur, Department of Basic Science and Humanity.
July, 2019 – Nov, 2019	■ Guest Faculty. Narula Institute of Technology, Agarpara, Department of Basic Science and Humanity.

M.Sc. Project

May, 2019	■ Project Title Influence Of The Fear Function In The Stability Of Prey-Predator Models. Submitted under the guidance of Dr. Shariful Alam at the Department of Mathematics, Indian Institute of Engineering Science and Technology, Shibpur, India.
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Research Publications

Journal Articles

- 1 B. Banik, H. Mondal, Y. Xing, and S. Alam, "Solution of first-order non-homogeneous pentagonal neutrosophic differential equation in bernoulli's form," *Neutrosophic Sets and Systems*, Accepted in 2024.
- 2 B. Banik, S. Alam, and A. Chakraborty, "Linear programming based extended copras technique for the highway project planning oriented mcgdm problem in cylindrical neutrosophic domain," *Operations Research Forum*, vol. 6, no. 84, pp. 1–48, 2025. 🔗 URL: <https://doi.org/10.1007/s43069-025-00481-6>.
- 3 B. Banik, S. Alam, and A. Chakraborty, "Analysis of economic setback of different countries due to covid-19 surge by advanced multi moora strategy under pentagonal neutrosophic realm," *Process Integration and Optimization for Sustainability*, vol. 8, pp. 975–991, 2024. 🔗 URL: <http://dx.doi.org/10.1007/s41660-024-00402-w>.
- 4 B. Banik, A. Chakraborty, A. Barman, and S. Alam, "Multi-method approach for new vehicle purchasing problem through mcgdm technique under cylindrical neutrosophic environment," *Soft Computing*, pp. 1–17, 2024. 🔗 URL: <http://dx.doi.org/10.1007/s00500-023-09520-y>.
- 5 B. Banik, S. Alam, and A. Chakraborty, "Comparative study between gra and merec technique on an agricultural-based mcgdm problem in pentagonal neutrosophic environment," *International Journal of Environmental Science and Technology*, vol. 20, no. 12, pp. 13 091–13 106, 2023. 🔗 URL: <http://dx.doi.org/10.1007/s13762-023-04768-1>.
- 6 B. Banik, S. Alam, and A. Chakraborty, "A novel integrated neutrosophic cosine operator based linear programming anp-edas mcgdm strategy to select anti-pegasus software," *International Journal of Information Technology & Decision Making*, vol. 23, no. 4, pp. 1633–1669, 2023. 🔗 URL: <https://doi.org/10.1142/S0219622023500529>.
- 7 A. Chakraborty, B. Banik, S. Broumi, and S. Salahshour, "Graded mean integral distance measure and vikor strategy based mcdm skill in trapezoidal neutrosophic number.," *International Journal of Neutrosophic Science (IJNS)*, vol. 18, no. 2, 2022. 🔗 URL: <http://dx.doi.org/10.54216/IJNS.180205>.
- 8 N. Singh, A. Chakraborty, B. Banik, S. D. Biswas, and M. Majumdar, "Digital banking chatbots related mcdm problem by todim strategy in pentagonal neutrosophic arena," *Journal of Neutrosophic and Fuzzy Systems*, vol. 4, no. 2, pp. 26–41, 2022. 🔗 URL: <http://dx.doi.org/10.54216/JNFS.040202>.

- 9 A. Chakraborty, B. Banik, S. P. Mondal, and S. Alam, "Arithmetic and geometric operators of pentagonal neutrosophic number and its application in mobile communication service based mcgdm problem," *Neutrosophic Sets and Systems*, vol. 32, no. 1, pp. 61–79, 2020.  URL: <http://dx.doi.org/10.5281/zenodo.3723145>.

Books and Chapters

- 1 B. Banik, A. Chakraborty, and S. Alam, "Data-driven modelling with fuzzy sets: A neutrosophic perspective," in CRC Press, 2024, ch. First Order Non-Homogeneous Ordinary Differential Equation with Initial Value Condition under Pentagonal Neutrosophic Number Environment.  URL: <http://dx.doi.org/10.1201/9781003487104>.

Communicated Research Articles

- 1 B. Banik, P. Garai, S. Alam, and A. Chakraborty, "A novel mcgdm model on dynamic covid-waves analysis in india under linguistic neutrosophic environment," Communicated to a SCI-indexed journal.
- 2 Y. Xing, Z. Li, J. Pan, and B. Banik, "An online doctor-patient two-sided matching decision-making method with multiple linguistic term sets considering the fairness," Communicated to a SCI-indexed journal.

Workshops and Conferences

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| 6-7 Mar, 2025 |  | Presented a paper in "International Conference on Mathematics and Applications-2025 (ICMA-2025)" at <i>Department of Mathematics, The University of Burdwan</i> . |
| 6-10 Jan, 2025 |  | Participation in the workshop "Understanding Supply Chain and its Research Perspectives using Python" at <i>SQC & OR Unit, Indian Statistical Institute, Kolkata</i> . |
| 6 Aug, 2021 |  | Participation in the national webinar "Usage of Mathematical Tools in MATLAB" at <i>Department of Mathematics, Krishna University, Machilipatnam</i> . |
| 22 June, 2020 |  | Participation in the national webinar "The role of Mathematical Modelling in COVID 19 Pandemic Response: Main Challenges" at <i>Department of Mathematics, Vijaygarh Jyotish Ray College, Kolkata</i> . |

Skills

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| Languages |  | Strong reading, writing and speaking competencies for English, Bengali, Hindi. |
| Coding |  | C, Matlab, Mathematica, $\text{\LaTeX}$. |
| Manuscript reviewing |  | Certified reviewer for journals such as Scientific Reports (Springer Nature), Evolutionary Intelligence (Springer Nature), Neutrosophic Sets and Systems etc. |
| Misc. |  | Academic research, teaching, training, project-assistantship, $\text{\LaTeX}$ typesetting and publishing. |

Miscellaneous Experience

Awards and Achievements

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| 2017 |  | IIT JAM |
| 2019 |  | GATE, conducted by the Department of Higher Education, Ministry of Human Resource Development, Govt. of India. |

Miscellaneous Experience (continued)

2021–Till date  **Institute Ph.D. fellowship**, Indian Institute of Engineering Science and Technology, Shibpur.

References

Prof Shariful Alam

Associate Professor

Indian Institute of Engineering Science and Technology, Shibpur, India.

Botanical Garden Area, Howrah, West Bengal 711103, India.

 salam50in@yahoo.co.in

 +91-9874452995

Prof Ujjal Debnath

Associate Professor

Indian Institute of Engineering Science and Technology, Shibpur, India.

Botanical Garden Area, Howrah, West Bengal 711103, India.

 ujjaldebnath@yahoo.com

 +91-9433353462

Prof Smita Pal Sarkar

Associate Professor

Indian Institute of Engineering Science and Technology, Shibpur, India.

Botanical Garden Area, Howrah, West Bengal 711103, India.

 smitasarkar.math@faculty.iiests.ac.in

 +91-7044055216/ +91-9433309939

Dr. G.S. Mahapatra

Associate Professor

National Institute of Technology Puducherry (NITPY), India.

Thiruvettakudy, Karaikal Puducherry Dist. Pincode: 609609

 gs.mahapatra@nitpy.ac.in

 +91-7598111189/ +91-6379838530